

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Work Integrated Learning Programmes Division

Cluster Programme - M. Tech in AI & ML

I Semester , 2023 – 24(January,2024)

Mid semester Examination (**Regular**)

Course No : AIMLC ZC418
Course Title : Introduction to Statistical Methods
Nature of Exam. : Closed Book (Online)
Weightage : 30 Marks
Duration : 120 minutes
Date : 21st January,2024_FN

Number of questions:4

Number of Pages: 4

Q. No	Question	Marks
Q.1.a)	Let A,B and C are three events such that $P(A) = 0.26, P(B) = 0.36, P(C) = 0.48, P(A \text{ and } B) = 0.03, P(B \text{ and } C) = 0.05, P(A \text{ and } C) = 0.04$ and $P(A \text{ and } B \text{ and } C) = 0.01$. Then find $P(A \text{ and } B^c), P(A^c \text{ and } B), P(A^c \cup B^c \cup C^c)$ and $P(A \text{ only}), P(B \text{ only})$ and $P(C \text{ only})$. (Note: Superscript c indicates the compliment of the corresponding event) Solution:	3 M

	$P(A) = 0.26$ $P(B) = 0.36$ $P(C) = 0.48 \rightarrow \text{[Tybo error should be } P(B) = 0.48]$ $P(A \cap B) = 0.03$ $P(B \cap C) = 0.05$ $P(A \cap C) = 0.04$ $P(A \cap B \cap C) = 0.01$ (i) $[P(A \cap B^c)] = P(A) - P(A \cap B) = 0.26 - 0.03$ $= 0.23$ <u>Ans</u> (ii) $P(A^c \cap B) = P(B) - P(A \cap B) = 0.36 - 0.03$ $= 0.33$ <u>Ans</u> (iii) $P(A \cup B \cup C) = 1 - P(A \cap B \cap C) = 1 - 0.01 = 0.99$ <u>Ans</u> (iv) $P(A \text{ only}) = P(A) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$ $= 0.26 - 0.03 - 0.04 + 0.01$ $= 0.2$ <u>Ans</u> (v) $P(B \text{ only}) = P(B) - P(A \cap B) - P(B \cap C) + P(A \cap B \cap C)$ $= 0.36 - 0.03 - 0.05 + 0.01$ $= 0.29$ <u>Ans</u> 118 $P(C \text{ only}) = P(C) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$ $= 0.48 - 0.04 - 0.05 + 0.01 = 0.4$ <u>Ans</u>	
Q.1.b)	Validate the statement "Mutually exclusive events are always independent". Validate with logic and examples in support. Solution: "Mutually exclusive events are always independent " is not necessarily true. Mutually exclusive events and independent events are distinct concepts in probability. Mutually Exclusive Event: In probability theory, two events are said to be mutually exclusive if they cannot occur at the same time or simultaneously. In other words, mutually exclusive events are called disjoint events. If two events are considered disjoint events, then the probability of both events occurring at the same time will be zero. If A and B are the two events, then the probability of disjoint of event A and B is written by: Probability of Disjoint (or) Mutually Exclusive Event = $P(A \text{ and } B) = 0$	4 M

	Independent Events: - Independent events are events for which the occurrence of one event does not affect the occurrence of the other. In other words, the probability of one event happening does not depend on whether the other event happens or not. - Mathematically, two events A and B are independent if $P(A \text{ and } B) = P(A) * P(B)$. Validation: - Mutually exclusive events can be independent, but they are not always independent. Example: - Consider two events: - Event A: Rolling a 1 on a six-sided die. - Event B: Rolling a 2 on the same six-sided die. - These events are mutually exclusive because rolling a 1 and rolling a 2 cannot happen simultaneously. However, they are also independent events because the occurrence of rolling a 1 does not affect the probability of rolling a 2, and vice versa. Counterexample: - Consider two events: - Event C: Selecting a red card from a standard deck of playing cards. - Event D: Selecting a face card (king, queen, or jack) from the same deck. - These events are mutually exclusive because a card cannot be both red and a face card. However, they are not independent, as the knowledge that a card is red changes the probability of it being a face card (since only half of the red cards are face cards). In conclusion, while mutually exclusive events can be independent in certain cases, they are not always independent. The independence of events depends on the specific probabilities associated with the events and their outcomes.	
Q.2.a)	Let A and B are two independent events such that $P(A) = 0.30$, $P(B) = 0.25$. Then find i) $P(B A \cup B)$ ii) $P(A A \cap B)$ iii) $P(A^c A \cup B)$	3 M

iv) $P(A^c | A \cap B)$

Solution:

(2Q) Sol: ^(3m)
(a) Given A and B are independent events.
 $P(A) = 0.30$, $P(B) = 0.25$

$$P(A \cap B) = P(A) \cdot P(B) \\ = 0.3 \times 0.25 = 0.075$$

$$(i) \quad P(B/A \cup B) = \frac{P[B \cap (A \cup B)]}{P(A \cup B)} = \frac{P(B)}{P(A \cup B)} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \rightarrow 1^m$$
$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ = 0.3 + 0.25 - 0.075 = 0.475$$

$$\therefore P(B/A \cup B) = \frac{0.25}{0.475} = \underline{0.526}$$

$$(ii) \quad P(A/A \cap B) = \frac{P(A \cap (A \cap B))}{P(A \cap B)} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} 1/2^m$$
$$= \frac{P(A \cap B)}{P(A \cap B)} = 1$$

$$(iii) \quad P(A^c/A \cup B) = \frac{P(A^c \cap (A \cup B))}{P(A \cup B)} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \rightarrow 1^m$$
$$= \frac{P(B) - P(A \cap B)}{P(A \cup B)}$$

	$= \frac{0.25 - 0.075}{0.475} = 0.3684$ $(iv) P(A^c/A \cap B) = \frac{P[A^c \cap (A \cap B)]}{P(A \cap B)} \left. \vphantom{\frac{P[A^c \cap (A \cap B)]}{P(A \cap B)}} \right\} \rightarrow 1/2^m$ $= 0$	
Q.2.b)	In a manufacturing company Machine 1 produces 40% of the items and Machine 2 and Machine 3 produces 25% and 35% of the items respectively. But from the past records it is found that 15%, 20% and 25% of the items they produce are defective. Then find i) Total percentage of defective produced ii) If a defective item is selected randomly, then find the probability that it is produced by Machine 1 iii) If a defective item is selected randomly, then find the probability that it is not produced by Machine 2 Solution:	4 M

2.b) sol: Given $P(M_1) = 0.40$, $P(M_2) = 0.25$, $P(M_3) = 0.35$ (4m)
 let D = defective item
 $P(D/M_1) = 0.15$
 $P(D/M_2) = 0.2$
 $P(D/M_3) = 0.25$
 (i) $P(D) = (0.4) \cdot (0.15) + (0.25) \cdot (0.2) + (0.35) \cdot (0.25)$
 $= 0.1975 \rightarrow 1m$
 $\therefore$ Total Percentage of defective produced is 19.75%
 (ii) $P(M_1/D) = \frac{P(M_1) \cdot P(D/M_1)}{P(D)}$
 $= \frac{(0.4) \cdot (0.15)}{0.1975} \rightarrow 1m$
 $= 0.3038$
 (iii) $P(M_2/D) = 1 - P(M_1/D) \rightarrow 1m$
 $P(M_2/D) = \frac{P(M_2) \cdot P(D/M_2)}{P(D)} = \frac{0.25 \times 0.2}{0.1975} = 0.2532$
 $\therefore P(M_2/D) = 1 - 0.2532$
 $= 0.7468$

Q.3.a)

Let $f(x) = k(x+1)/2$ where $0 < x < 2$,
 $= 0$, otherwise

Then find

- k value
- $P(1/2 < x < 2)$
- $P(X > 1/4)$

Solution:

The function is given by:

$$f(x) = \frac{k(x+1)}{2} \text{ for } 0 < x < 2$$

$$f(x) = 0 \text{ otherwise}$$

i) To find Value of k:

$$\int_0^2 \frac{k(x+1)}{2} dx = 1$$

$$\frac{k}{2} \left[\frac{x^2}{2} + x \right]_0^2 = 1$$

$$\frac{k}{2} [(2+2) - (0+0)] = 1$$

$$\text{Therefore } k = \frac{1}{2}$$

4 M

ii) **P(1/2 < x < 2)**

$$P\left(\frac{1}{2} < x < 2\right) = \int_{1/2}^2 \frac{1}{2} \cdot \frac{(x+1)}{2} dx$$

$$P\left(\frac{1}{2} < x < 2\right) = \frac{1}{4} \left[\frac{x^2}{2} + x \right]_{\frac{1}{2}}^2$$

$$= \frac{27}{32} = 0.84375$$

iii) **P(X > 1/4)**

$$P(X > 1/4) = \frac{1}{4} \left[\frac{x^2}{2} + x \right]_{1/4}^2$$

$$= \frac{119}{128} = 0.9297$$

Q.3.b).

Consider the following probability distribution of X and Y.

4 M

Y	X			
	0	1	2	3
0	0.05	0.25	0.05	0
1	0.05	0.15	0.05	0.05
2	0.10	0.15	0.10	0

i). Find marginal distribution of X

ii). Find marginal distribution of Y

iii). Find P(X < 2 / Y < 2)

iv). Find P(X < 2 / Y = 1)

Solution:

i). Find marginal distribution of X

X	0	1	2	3	Total
P(x)	0.2	0.55	0.2	0.05	1

ii). Find marginal distribution of Y

Y	0	1	2	Total
P(y)	0.35	0.3	0.35	1

iii). Find P(X < 2 / Y < 2)

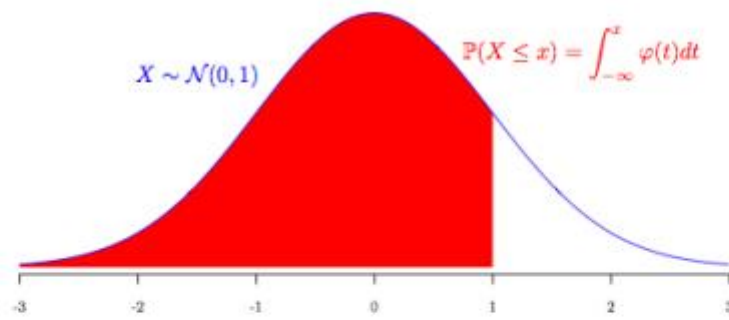
$$P(X < 2 | Y < 2) = \frac{P(X < 2 \cap Y < 2)}{P(Y < 2)}$$

$$= \frac{P(X = 0, Y = 0) + P(X = 1, Y = 0) + P(X = 0, Y = 1) + P(X = 1, Y = 1)}{P(Y = 0) + P(Y = 1)}$$

$$= \frac{0.05 + 0.05 + 0.25 + 0.15}{0.35 + 0.3} = \frac{10}{13} = 0.7692$$

	iv). Find $P(X < 2 \mid Y = 1)$ $P(X < 2 \mid Y = 1) = \frac{P(X < 2 \cap Y = 1)}{P(Y = 1)}$ $= \frac{P(X = 0, Y = 1) + P(X = 1, Y = 1)}{P(Y = 1)}$ $= \frac{0.05 + 0.15}{0.3} = \frac{2}{3} = 0.67$	
Q.4.a)	Consider a random variable X that follows binomial distribution with $n = 250$ and $p = 0.20$. Then find the probabilities that i). $X > 180$ ii). $X = 150$ iii). $150 < X < 180$ Ans: Here $np = 250 \times 0.2 = 50$ and $nq = 250 \times 0.8 = 200$, both > 15 Then normal approximation to binomial can be used with continuity correction using mean $= np = 50$ and $Sd = \sqrt{npq} = 6.32$ $Z = (x - \text{mean})/sd = (x - 50)/6.32$ (1) i) $P(X > 180) = P(X > 179.5) = P(Z > 20.49) = 0$ (1) ii) $P(X = 150) = P(149.5 < X < 150.5) = P(15.7 < Z < 15.9) = 0$ (1) iii) $P(150 < X < 180) = P(149.5 < X < 180.5) = P(15.7 < Z < 20.6) = 0$ (1)	4 M
Q.4.b)	Samples of size 45 are selected randomly from a population having mean 25 and standard deviation 4. Let X be a random variable associated with sampling distribution of means. Then find the probability that i). X lies between 20 and 30. ii). X is greater than 28 iv) X is less than 28 Solution: $Z = (x - \text{mean})\sqrt{n}/sd = (x - 25)\sqrt{45}/4 = (x - 25)1.67$ (1) i) $P(20 < X < 30) = P(-8.35 < Z < 8.35) = 1$ (1) ii) $P(X > 28) = P(Z > 5.01) = 0$ (1) iii) $P(X < 28) = P(Z < 5.01) = 1$ (1)	4 M

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414
-0.1	.46017	.45620	.45224	.44828	.44433	.44034	.43640	.43251	.42858	.42465
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-1	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08692	.08534	.08379	.08226
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-2	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-3	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-4	.00003	.00003	.00003	.00003	.00003	.00003	.00002	.00002	.00002	.00002



	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990